

How Are Continuous Motor Skills Learned and Represented

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Abstract

Most real-world motor actions require skilled continuous coordination. Here, we investigated how continuous motor performance improves with practice and how learned skills are represented. We recruited a large sample of participants ($N = 300$) to perform a tracing task, systematically varying how far ahead participants could preview the trajectory (5%–100%) to control the extent of advanced planning. To dissociate general skill learning from trajectory-specific learning, participants repeatedly practiced a single trajectory while interleaved random trajectories served as a baseline. To probe whether learning benefited from the trajectory's global structure, we introduced rotated versions of the trained trajectory. With practice, participants became faster and more accurate. These improvements reflected general gains that were similar across preview conditions, without evidence for an expanded planning horizon, alongside trajectory-specific gains that diminished as preview increased. When probed, we found transfer to rotated trajectories, indicating learning benefits from a global representation of trajectory structure. Together, these findings suggest that continuous skill learning reflects modest general improvements in planning and execution, but substantial trajectory-specific gains that benefit from improved predictions of upcoming movements grounded in the trajectory's global structure.

Keywords: motor learning, motor skill, motor planning, generalization, representation

Introduction

From calligraphy to Formula 1 racing, real-world motor skills require continuous actions that are planned and executed with both speed and precision - proficiency forged through practice. Prior work on discrete or sequential actions (e.g., reaching or keypress sequences) observed learning involves multiple process: (i) general execution and planning improvements (Ariani et al., 2021; Bashford et al., 2022), (ii) sequence-specific gains that arise from improved predictions of upcoming elements (Kashefi et al., 2024, 2025). However, whether these processes extend to continuous, real-world skills learning remains unclear.

To fill this gap, we asked two questions. First, how does continuous motor performance improve with practice? Building on the previous literature, we hypothesize that improvements also reflect general and trajectory-specific improvements in planning and execution. Second, how are learned motor skills represented? We

hypothesize that learning leverages a representation of global structure (i.e., the continuous trajectory's shape), not merely a sequence of individual movements.

Methods

We recruited a large sample of participants ($N = 300$) via Prolific to complete a web-based tracing task. On each trial, participants traced a 2D trajectory with a cursor as quickly and accurately as possible (Fig. 1A). Participants were randomly assigned to one of five preview conditions ($n = 60$ /group), which manipulated how much of the upcoming trajectory was visible (Fig. 1B). In the full-preview condition (100%), the entire path was shown; in reduced-preview conditions (5%, 10%, 25%, 50%), only a limited segment ahead of the cursor was visible, akin to navigating with a flashlight in the dark.

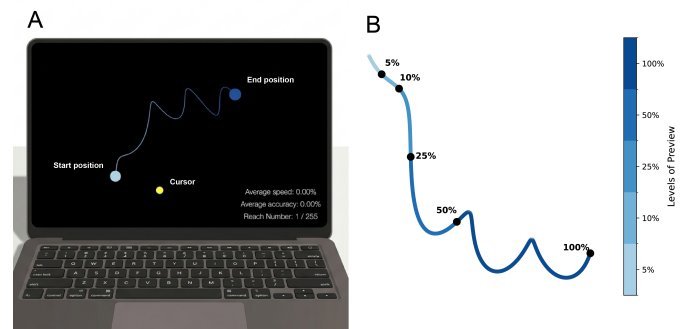


Figure 1: Web-based tracing task. (A) Task setup. (B) Preview manipulation of the trained trajectory.

The experiment consisted of 45 cycles (240 trials). Each cycle included four repetitions of the trained trajectory (180 trials), one random trajectory (45 trials), and rotated versions of the trained trajectory (presented with one-third probability across cycles; 15 trials). Trajectories were generated using a sum-of-sines algorithm (amplitude: 0.5-1.5; frequency: 0.4-0.6 (Yang et al., 2021), with all paths matched in length

Performance was quantified using movement time and spatial error. Movement time was defined as the interval from leaving the start position to reaching the endpoint. Error was computed as the mean Euclidean distance between the participant's path and the target trajectory, resampled to 200 evenly spaced points (reported in arbitrary units). Early learning was defined as the first 15 cycles, and late learning as the last 15 cycles.

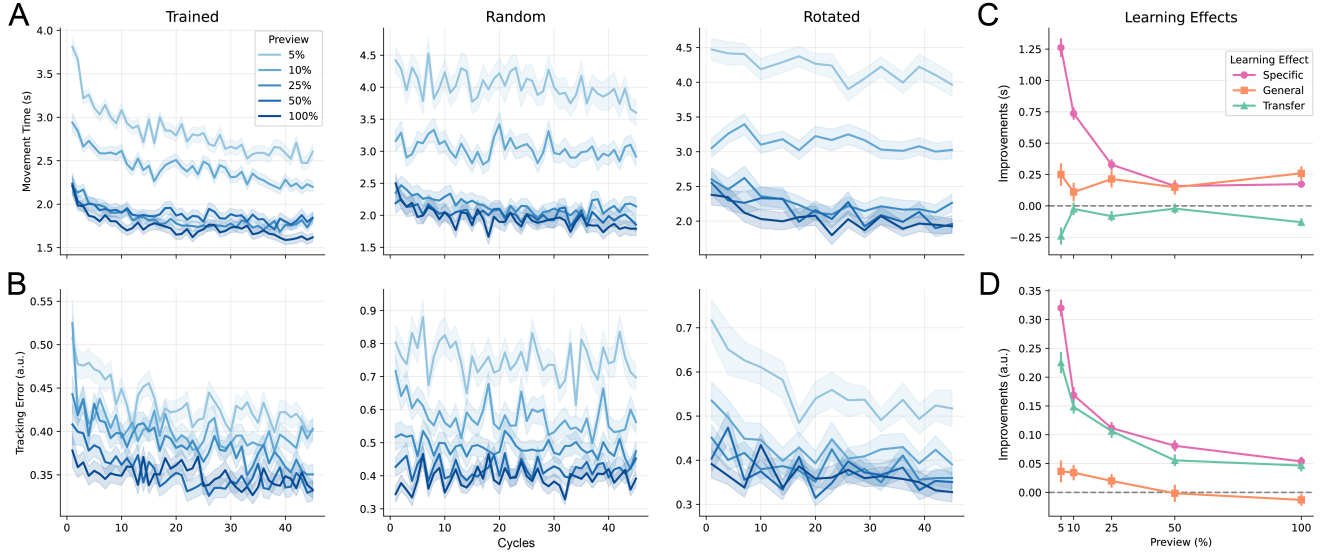


Figure 2: Motor skill learning across preview conditions and trajectory types. Preview ranged from 5% to 100% (blue shading), with trajectory types shown across columns. Practice improved (A) movement time and (B) spatial error. (C–D) General, trajectory-specific, and transfer effects for movement time and error.

Results

General skill improvements reflect increased continuous motor efficiency rather than expanded planning capacity. We first asked whether motor skill is constrained by planning. To test this, we analyzed baseline motor performance in trials with minimal learning (early random trials). We found limited preview led to markedly slower movement time ($F(4, 281) = 64.7, p < .001$) and larger errors ($F(4, 281) = 68.2, p < .001$) whereas longer preview yielded better performance. We next quantified the "planning horizon", the preview level beyond which additional trajectory information yielded minimal benefit by fitting an exponential function and identifying its asymptote. Planning horizon at this baseline stage was 31.7% (95% CI : 24.6–41.7), indicating the motor performance is constrained by a limited planning window.

We then asked whether general movement improved with practice. We compared early and late performance on random trajectories. With practice, participants move faster ($F(1, 281) = 40.1, p < .001$) and reduced error ($F(1, 281) = 6.0, p = .015$) across all preview conditions. Surprisingly, the planning horizon did not extend with learning (95% CI : 28.2–47.9; Early-Late = 4.6%, 95% CI : –0.7–12.0). Thus, general skill improvements reflect increase motor efficiency rather than an expansion of planning capacity.

Trajectory-specific gains reflect improved anticipation of upcoming movements. To qualify trajectory-specific learning, we compared late performance on trained versus random trajectories. With learning,

movement time were substantially faster ($F(1, 281) = 614.4, p < .001$) and error was markedly lower ($F(1, 281) = 815.5, p < .001$) for trained trajectories. These gains in speed ($F(4, 281) = 94.0, p < .001$) and accuracy ($F(4, 281) = 84.5, p < .001$) were strongly modulated by preview. Overall, trajectory-specific gains far exceeded general skill learning.

Learned continuous skills benefit from a global representation of trajectory structure. We then asked whether learned skills benefit from a representation of the global trajectory structure. To test this, we compared late performance on rotated against random trajectories. Across preview conditions, accuracy was higher for rotated than random trajectories ($t(285) = -16.9, p < .001$); however, movement times effects were more variable: some conditions show no change (10% and 50%, $t(285) = 0.7, p = 0.5$), whereas others were slower (5%, 25%, 100%, $t(285) = 5.3, p < .001$). Taken together, these results suggest that learned skills leverage the global structure of a trajectory to improve spatial accuracy, while at times shifting the speed–accuracy trade-off.

Conclusion

Continuous skill learning reflects modest general improvements in planning and execution but substantial trajectory-specific gains that benefit from improved predictions of upcoming movements grounded in the trajectory's global structure. Together, these findings extend principles derived from discrete, sequential learning to real-world, continuous motor skills.

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